My First STEM Coloring Book

Four Topics and Conversation Guide Sample

By Brandon Hui | Ages 5-7

Explore one topic from each STEM subject:

Science: Planets (book page 7)

Technology: Light Bulb (book page 27)

Engineering: Airplane (book page 57)

Mathematics: Patterns (book page 75)

Also included: the Science page from the Parent and Teacher Conversation Guide (book page 83), including the Planets entry.

This six-page preview contains selected pages, not the complete book.

Original book page numbers are retained.

For single-sided coloring, print each topic on a separate sheet.

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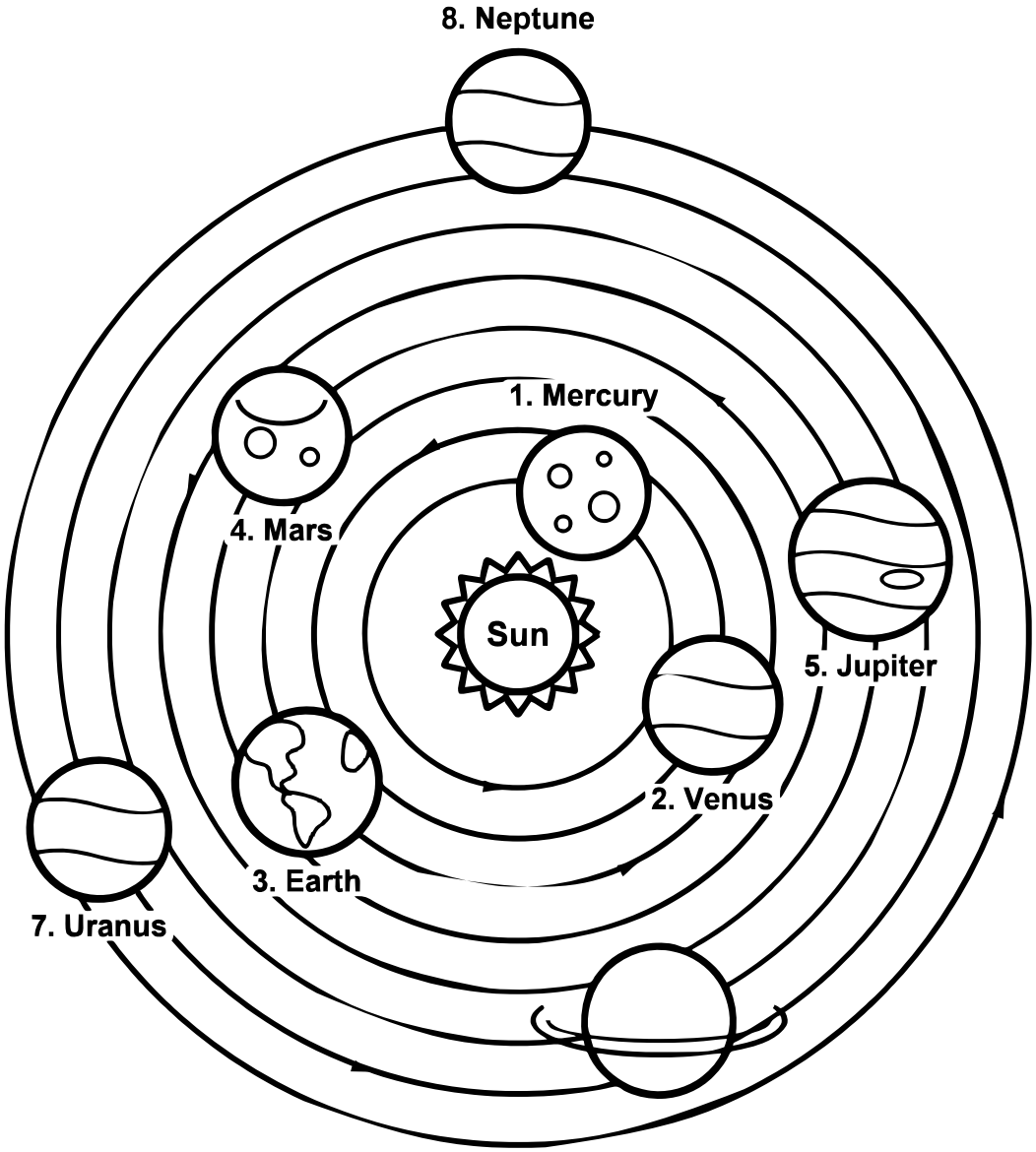
Illustrations generated with AI tools and reviewed by the author.

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Project: github.com/culturelight/my-first-stem-coloring-book

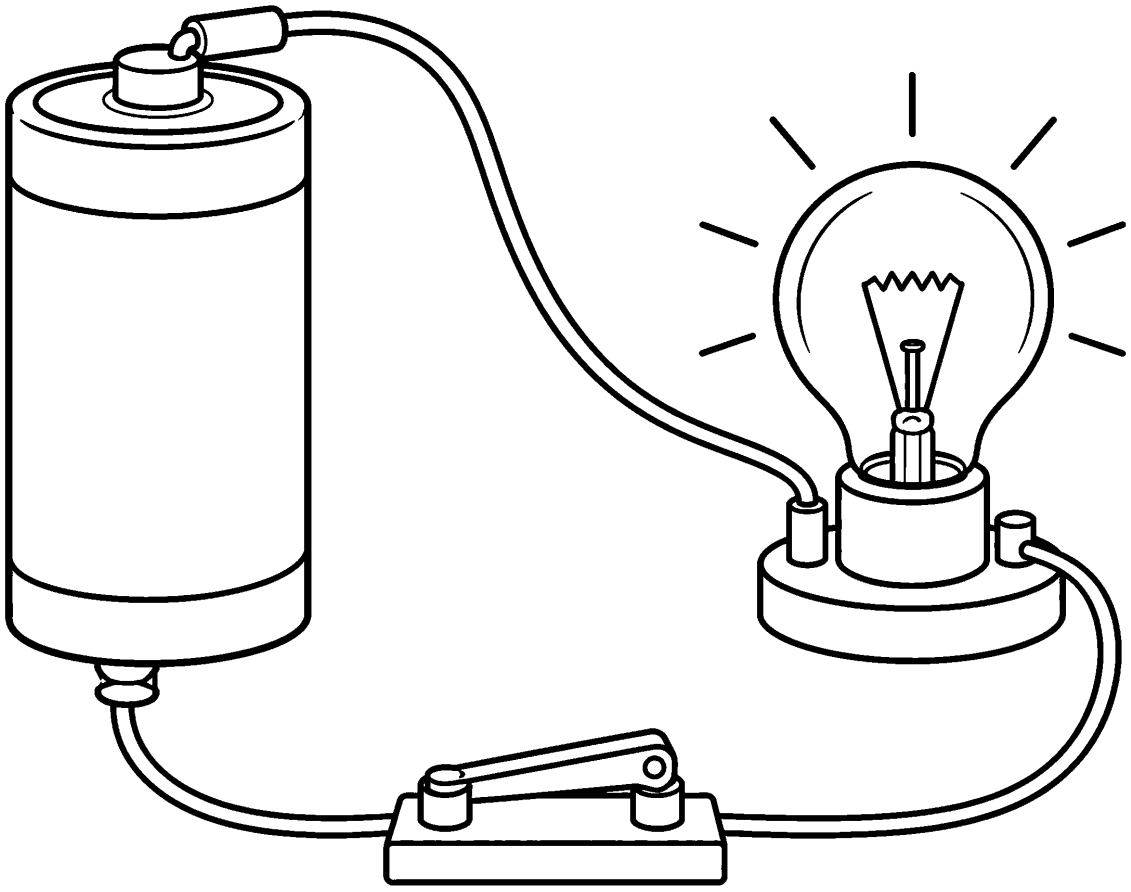
Planets



Sizes and distances are not to scale.
Follow each path around the Sun.

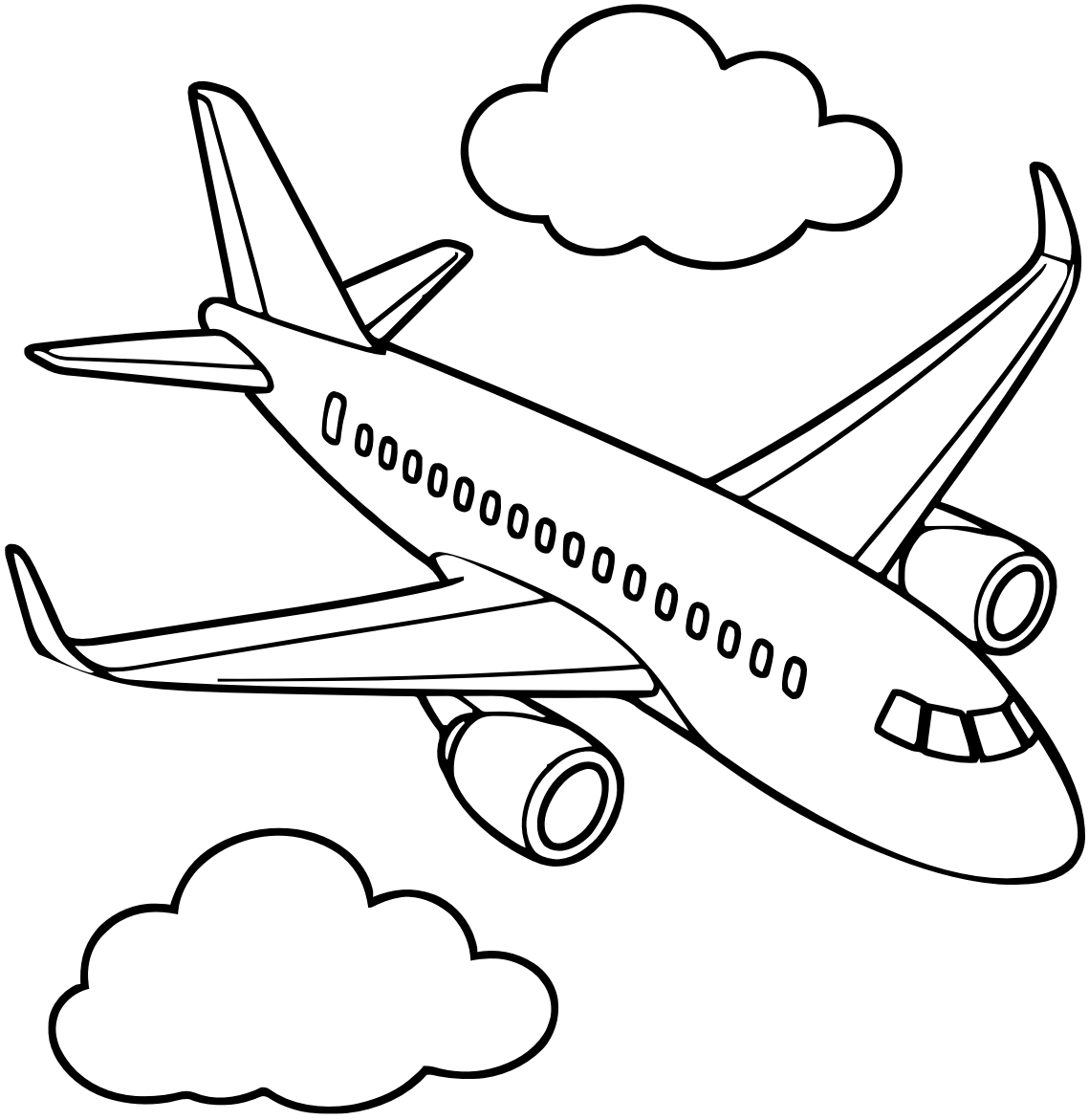
Eight planets travel around the Sun.
Earth is the planet we live on.

Light Bulb



A battery supplies electricity. When the switch closes the circuit, electricity can flow and light the bulb.

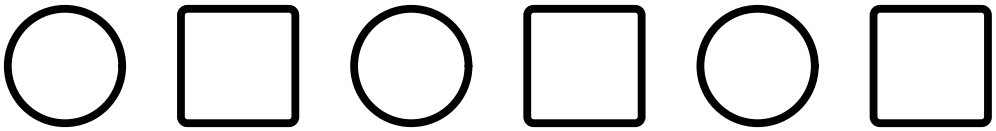
Airplane



As an airplane moves, air flows around its wings. The wings create lift that helps it stay in the air.

Patterns

AB



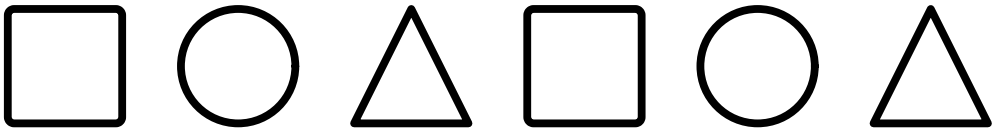
A B A B A B ?

AAB



A A B A A B ?

ABC



A B C A B C ?

A pattern follows a rule that repeats.
Look at each pattern. Draw the shape that comes next.

Parent and Teacher Conversation Guide

Read together, then choose a question. Let your child point, talk, or draw. Discoveries grow through many people sharing ideas. Your questions can start something new, too!

Science

Planets Page 7

Planets are worlds that orbit a star. People have watched them since ancient times; William Herschel discovered Uranus in 1781. Exploring planets helps us understand our home, Earth.

Which planet in the picture is Earth? Which planet is closest to the Sun?

The planets are shown in order from the Sun; their sizes and distances are not to scale.

Plant Page 9

Plants are living things that usually make food using sunlight. Farmers and nature watchers have studied them for thousands of years. Their shared knowledge helps us grow food and care for habitats.

Which parts of the plant can you name? What do you think would happen if it had no water?

Look at a real plant together and find its roots, stem, leaves, or flowers.

Five Senses Page 11

Senses help our bodies gather information. People have explored sight, hearing, smell, taste, and touch since ancient times; many scientists added discoveries. Understanding senses helps us learn and care for one another.

What can you see and hear right now? How could you describe an orange using more than one sense?

Taste only familiar food with a grown-up; never taste unknown objects or plants.

Magnet Page 13

Magnets attract iron and some other materials. People knew natural magnets in ancient times; William Gilbert explained Earth as a giant magnet in 1600. Magnets help compasses guide travelers and motors move.

Which objects in the picture might a magnet pull? Would you expect it to pull a wooden spoon?

Not every metal is attracted to a magnet. Compare predictions using a large classroom magnet.

Sound Page 15

Sound is vibration traveling through matter, such as air. Ancient musicians explored it long before modern scientists measured it. This shared learning helps people make music, communicate, and design hearing aids.

What happens when you tap a drum gently? How could you make a louder or quieter sound?

Vibration means a repeated back-and-forth movement. Listen at a comfortable volume.

Volcano Page 17

Volcanoes are openings where hot rock, ash, and gases escape. People have observed eruptions since ancient times; no single person discovered them. Scientists study volcanoes to help communities prepare and stay safer.

Where is the lava coming out? How is lava different from the rocks around the volcano?

Hot melted rock is called magma underground and lava after it reaches the surface.

Water Cycle Page 19

The water cycle is water moving between land, air, and oceans. People have studied rain and rivers since ancient times; many observers helped explain the cycle. Understanding it helps us protect water supplies.

Can you trace the water's journey using the labels? Where might a puddle's water go on a sunny day?

Evaporation: liquid becomes vapor. Condensation: vapor becomes droplets. Precipitation: water falls. Collection: water gathers.

Microscope Page 21

A microscope makes tiny details look bigger. In 1665, Robert Hooke published drawings of cork and named its tiny spaces cells. Microscopes help scientists study living things and understand disease.

What might a leaf look like through a microscope? Why might a scientist want to see tiny details?

A magnifying glass is a familiar starting point for talking about magnification.

Science Lab Page 23

A laboratory is a place to test ideas. Labs grew over centuries; Marie and Pierre Curie discovered polonium in 1898 through careful experiments. Lab discoveries help people develop medicines and useful materials.

What changes can you see in the picture? What would you watch for if two materials were mixed?

Observation means noticing carefully. Use only grown-up-approved materials for any activity.

History and science: NASA/JPL; Royal Society; Science Museum; Nobel Prize; USGS.